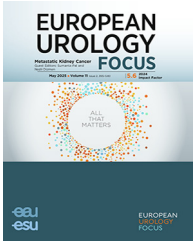
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Grey Zone

KIM-1 as a Prognostic Marker in Renal Cell Carcinoma

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1. Introduction

There are currently no cancer biomarkers routinely used to guide clinical management of renal cell carcinoma (RCC). Kidney injury molecule-1 (KIM-1), a transmembrane glycoprotein, is overexpressed by cancers that originate from the proximal tubular epithelium, including clear-cell RCC (ccRCC) and papillary RCC. The KIM-1 extracellular domain is detectable in as little as 25 µl of urine and plasma, making this protein a potential minimally invasive biomarker for RCC [1,2]. KIM-1 is being evaluated as a clinical biomarker for RCC in different disease states including localized and metastatic disease (Fig. 1).

2. KIM-1 before nephrectomy

Higher circulating KIM-1 levels are associated with a higher risk of RCC diagnosis. In a nested case-control study of 190 RCC cases and 190 noncancer control subjects, patients with newly diagnosed RCC had elevated KIM-1 levels in banked plasma samples up to 5 yr before diagnosis. The RCC incidence rate ratio for a doubling in KIM-1 was 1.71 (95% confidence interval [CI] 1.44–2.03; $p < 0.0001$) [2].

KIM-1 has also been investigated as a risk stratification tool among patients with suspicious renal masses. In a nested case-control study involving 162 case-control pairs, plasma KIM-1 distinguished ccRCC from benign masses, with an area under the receiver operating curve (AUC) of 0.81 (95% CI 0.76–0.86). While plasma KIM-1 was elevated in cases with large and small ccRCCs, its discrimination was better among patients with larger tumors [3]. In an independent cohort of 247 cT1a small renal masses, circulating KIM-1 distinguished between benign and malignant tumors, with an AUC of 0.73 (95% CI 0.64–0.82) [3]. Multivariable analysis revealed that patients with ccRCC and higher circulating KIM-1 had worse metastasis-free survival (hazard ratio [HR] per unit increase in log KIM-1: 1.29, 95% CI 1.10–1.53) and overall survival (OS; HR per unit increase in log KIM-1: 1.31, 95% CI 1.10–1.5).

3. Adjuvant setting

Data from several trials have demonstrated the potential utility of KIM-1 as a biomarker after RCC resection. In the phase 3 ECOG-ACRIN 2805 (ASSURE) trial evaluating adjuvant sunitinib, sorafenib, or placebo among patients with high-risk RCC, higher KIM-1 after nephrectomy was associated with worse disease-free survival (DFS) and OS in all study arms ($p < 0.001$), suggesting that circulating KIM-1 could be used as a marker for microscopic residual disease [4]. Furthermore, addition of postnephrectomy KIM-1 levels improved existing clinical prognostic models (Stage, Size, Grade, and Necrosis [SSIGN] concordance 0.57 vs 0.43, $p = 0.05$; UCLA International Staging System [UISS] concordance 0.60 vs 0.40, $p = 0.0005$) [4].

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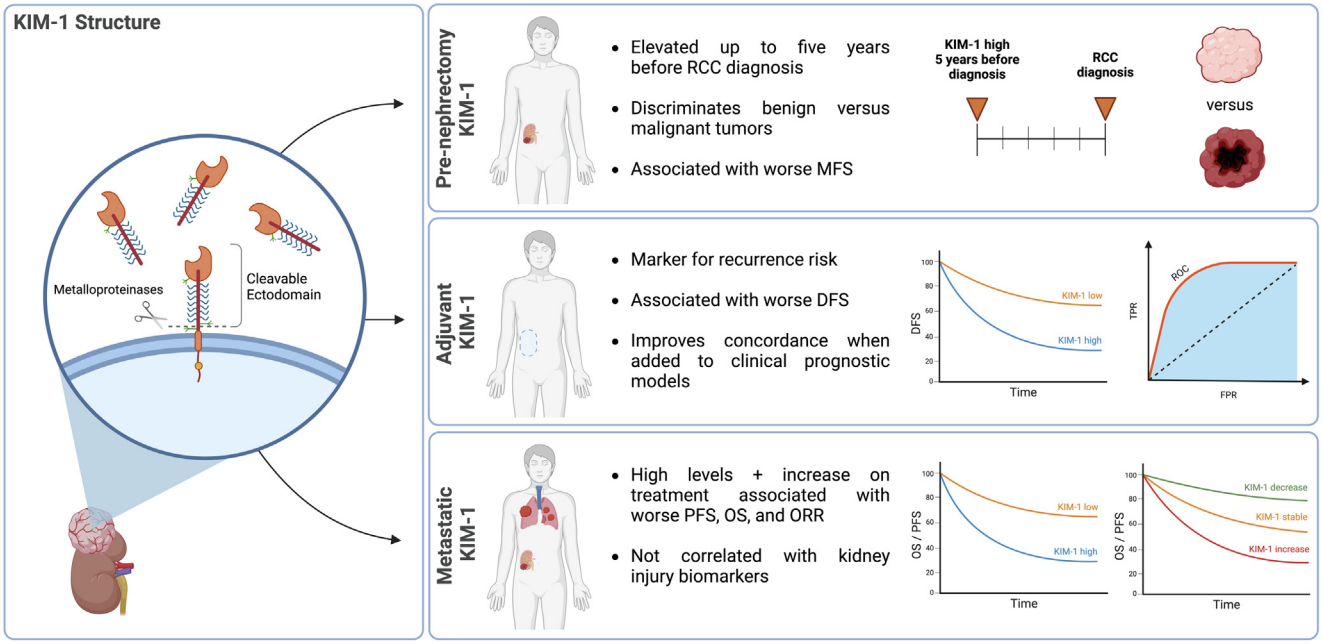


Fig. 1 – Rationale and clinical data supporting circulating KIM-1 as a biomarker for renal cell carcinoma.

The role of KIM-1 in predicting adjuvant response is further supported by a post hoc analysis of the CheckMate 914 Part A trial, which evaluated ipilimumab plus nivolumab versus placebo in the adjuvant setting. Ipilimumab and nivolumab had no DFS benefit in the intention-to-treat (ITT) population. However, the highest quartile for pretreatment serum KIM-1, versus all other quartiles, was associated with worse DFS and a trend towards a DFS benefit from adjuvant nivolumab plus ipilimumab (HR 0.60, 95% CI 0.34–1.04) [5].

In the IMmotion010 trial, patients with resected RCC and higher risk of recurrence received adjuvant atezolizumab versus placebo. A multiplexed proteomic assay revealed that KIM-1 was the most differentially enriched protein in serum at recurrence in comparison to postoperative baseline after nephrectomy. Furthermore, patients with high baseline KIM-1 levels in serum had shorter DFS (HR 1.75, 95% CI 1.40–2.17). While there was no benefit found for adjuvant atezolizumab in the ITT population, the subgroup of patients with high KIM-1 at baseline had longer median DFS with atezolizumab in comparison to placebo (not evaluable vs 21.16 mo; HR 0.72, 95% CI 0.52–0.99) [6].

4. Metastatic setting

KIM-1 has been associated with prognosis and response to treatment in metastatic (m)RCC. In the CheckMate 009 trial, patients with ccRCC were treated with nivolumab every 3 wk. Higher KIM-1 at baseline was associated with worse OS (univariable $p = 0.003$), and any decrease in KIM-1 at week 3 (after 1 cycle of nivolumab) was associated with better progression-free survival (PFS; HR 0.22, 95% CI 0.097–0.50) after multivariable adjustment [7].

The role of KIM-1 in monitoring patients on immunotherapy is further supported by a post hoc analysis of CheckMate 214, which investigated nivolumab plus ipilimumab versus sunitinib in first-line mRCC. Among 821 patients across both arms, high KIM-1 at baseline was associated with worse clinical outcomes, with shorter OS independent of International Metastatic RCC Database Consortium risk group, nephrectomy status, and tumor burden. Furthermore, changes in KIM-1 from baseline to 3 wk (>30% decrease vs >30% increase) were strongly associated with PFS, OS, and the overall response rate (ORR) among patients treated with nivolumab plus ipilimumab (median PFS: 70.8 vs 4.3 mo; median OS: 85.4 vs 26.6 mo; ORR 69.3% vs 13.9%). Early 3-wk changes in KIM-1 were not associated with treatment outcomes after sunitinib [8].

KIM-1 was evaluated in a post hoc analysis of the JAVELIN Renal 101 trial of immune checkpoint inhibitor plus VEGFR-targeted therapy (avelumab + axitinib) versus sunitinib in advanced RCC. Lower circulating KIM-1 at baseline was associated with longer PFS and OS in both groups. KIM-1 levels at baseline were significantly lower in the responder group versus the nonresponder group ($p = 0.01$), and in the exceptional responder subgroup versus the intermediate responder subgroup ($p = 0.0015$). In paired analyses of sequential measurements, circulating KIM-1 decreased from baseline to 12 wk ($p < 0.0001$) in both arms, and the decrease in KIM-1 was associated with longer PFS and OS [9]. Similarly, a post hoc analysis of the Tide-A trial of avelumab plus intermittent axitinib showed that high baseline KIM-1 was associated with worse OS ($p = 0.0019$) [10].

Although KIM-1 is overexpressed in kidney injury, available data suggest that concurrent kidney injury is not a confounder for the association between KIM-1 and RCC outcomes. Among prediagnostic samples [2], KIM-1 was not correlated with kidney injury markers (TNFR1, TNFR2, creatinine) and its association with RCC risk was not impacted by addition of these markers to the model. Similar findings were seen in the metastatic setting [11]. These data suggest that among patients with RCC, the majority of circu-

lating KIM-1 comes from RCC cells rather than injured kidney epithelium, even in the presence of coincident renal injury.

5. Future directions

In order for circulating KIM-1 to be useful in clinical practice, prospective validation is needed. This will require the inclusion of prespecified KIM-1 hypotheses in randomized clinical trial designs. Such trials would determine whether use of KIM-1 measurements to guide treatment decisions can improve clinical outcomes.

Currently, guidelines from the European Association of Urology and National Comprehensive Cancer Network recommend partial nephrectomy as the preferred option for the management of cT1a RCC. However, retrospective long-term follow-up data have shown that patients with cT1a tumors and low KIM-1 levels did not develop metastases, while patients with cT1b tumors and low KIM-1 levels had minimal metastatic potential [3]. Therefore, future studies could investigate combining radiology with KIM-1 measurement for renal mass surveillance. As one potential trial design, patients with cT1a tumors and low baseline KIM-1 could be randomized to immediate partial nephrectomy versus active surveillance, with a prespecified plan for intervention triggered by an increase in tumor size or KIM-1 levels. Such trials would show whether (1) low KIM-1 at baseline can be used to select renal masses for active surveillance and (2) serial KIM-1 levels can complement imaging for surveillance of patients with renal masses.

Additional clinical questions for prospective validation include (1) whether patients with high circulating KIM-1 after RCC resection benefit from intensified adjuvant immunotherapy; (2) whether patients with low circulating KIM-1 after RCC resection can safely forgo adjuvant immunotherapy; (3) whether patients with high circulating KIM-1 levels but no radiologic confirmation of disease will benefit from early systemic therapy; and (4) whether patients receiving immunotherapy who have a suboptimal KIM-1 decrease at 3 wk will benefit from a change in treatment. The establishment of KIM-1 risk cutoffs specific to each RCC disease stage should also be a priority.

To date, retrospective studies have used a variety of techniques to evaluate circulating KIM-1, including antibody-based microbead assays, electrochemiluminescence assays, and aptamer-based assays [2–7,11]. Regardless of the assay used, circulating KIM-1 has been associated with RCC outcomes. However, the distribution of KIM-1 differs between assays, so future clinical KIM-1 assays should be standardized to avoid the generation of multiple different cutoffs.

Besides KIM-1, other emerging minimally invasive biomarkers for RCC are also under investigation. These include circulating tumor DNA assays, carbonic anhydrase IX–based positron emission tomography imaging, and epigenetic cell-free DNA assays [12–15]. In the future, integration of circulating KIM-1 with these other emerging biomarkers might further improve diagnostic accuracy.

6. Conclusions

There is an unmet need for a minimally invasive biomarker in RCC. KIM-1 appears to be a promising candidate biomarker in multiple clinical scenarios. Additional validation in prospective trials is needed to establish specific thresholds for implementation of KIM-1-based decision-making in clinical practice.

Conflicts of interest: David F. McDermott reports a consulting or advisory role for Bristol-Myers Squibb, Merck, Genentech/Roche, Pfizer, Exelixis, Novartis, Array BioPharma, Peloton Therapeutics, EMD Serono, Jounce Therapeutics, Alkermes, Lilly, Eisai, Calithera Biosciences, Iovance Biotherapeutics, Werewolf Therapeutics, Synthekine, AVEO Pharmaceticals, Xilio Therapeutics, and Cullinan Oncology; and institutional research funding from Bristol-Myers Squibb, Merck, Genentech, Novartis, and Alkermes. Toni K. Choueiri reports stock and other ownership interests in Precede Bio, Osel, Tempest Therapeutics, Pionyr, Curesponse, InnDura, and Primium; honoraria from HiberCell, Pfizer, Bayer, Novartis, GlaxoSmithKline, Merck, Bristol-Myers Squibb, Roche/Genentech, Eisai, Foundation Medicine, AstraZeneca, Exelixis, Prometheus, Ipsen, Sanofi/Aventis, Peloton Therapeutics, UpToDate, National Comprehensive Cancer Network (NCCN), Michael J. Hennessy Associates, Analysis Group, Clinical Care Options, PlatformQ Health, Navinata Health, Harborside Press, American Society of Clinical Oncology (ASCO), New England Journal of Medicine, Lancet Oncology, EMD Serono, Lilly, Tempest Therapeutics, Arcus Biosciences, Arcus Biosciences, Alkermes, Gilead Sciences, Scholar Rock, Janssen Oncology, Precede Bio, Aravive, Infinity Pharmaceuticals, European Society for Medical Oncology (ESMO), NiKang Therapeutics, Kanaph Therapeutics, and Gilead Sciences; a consulting or advisory role for Pfizer, Bayer, Novartis, GlaxoSmithKline, Merck, Bristol-Myers Squibb, Roche/Genentech, Eisai, Foundation Medicine, AstraZeneca, Exelixis, Prometheus, Ipsen, Sanofi/Aventis, Peloton Therapeutics, UpToDate, National Comprehensive Cancer Network (NCCN), Michael J. Hennessy Associates, Analysis Group, Clinical Care Options, PlatformQ Health, Navinata Health, Harborside Press, ASCO, New England Journal of Medicine, Lancet Oncology, EMD Serono, Lilly, Tempest Therapeutics, Arcus Biosciences, alkermes, Gilead Sciences, Scholar Rock, Janssen Oncology, Precede Bio, Aravive, Infinity Pharmaceuticals, ESMO, NiKang Therapeutics, Kanaph Therapeutics, Gilead Sciences, and Neomorph; institutional research funding from Pfizer, Novartis, Merck, Exelixis, TRACON Pharma, GlaxoSmithKline, Bristol-Myers Squibb, AstraZeneca, Peloton Therapeutics, Roche/Genentech, Agensys, Eisai, Takeda, Ipsen, Seattle Genetics/Astellas, Bayer, Roche, Calithera Biosciences, NiKang Therapeutics, Arcus Biosciences, and AVEO Pharmaceticals; intellectual property interests in patents for biomarkers of clinical response and benefit to immune checkpoint inhibitor therapy (PCT/US2018/058430; institutional), PBRM1 biomarkers predictive of anti-immune checkpoint response (PCT/US2018/12209; institutional), and circulationg tumor DNA technologies; travel and accommodation expenses from Pfizer, Bayer, Novartis, GlaxoSmithKline, Merck, Bristol-Myers Squibb, Roche/Genentech, Eisai, Foundation Medicine, Cerulean Pharma, AstraZeneca, Exelixis, Prometheus, Allegiant, Ipsen, Corvus Pharmaceuticals, Lpath, Alexion Pharmaceuticals, Sanofi/Aventis, UpToDate, Peloton Therapeutics, NCCN, Michael J. Hennessy Associates, Analysis Group, Kidney Cancer Association, Clinical Care Options, PlatformQ Health, Harborside Press, Navinata Health, New England Journal of Medicine, Lancet Oncology, EMD Serono, Heron, Lilly, and ESMO; and support for medical writing and editorial assistance from communications companies funded by pharmaceutical companies such as ClinicalThinking, Health Interactions, Envision Pharma Group, Fishawack Group of Companies, and Parexel. Joseph V. Bonventre reports stock and other ownership interests in MediBeacon, Autonomous Medical Devices, Innoviva, DxNow, Oisin, Pacific Biosciences Research Center, and Renalytix AI PLC; honoraria from the University of Washington; a consulting or advisory role for Sarepta Therapeutics, Praxis Therapeutics, Praxis Therapeutics, Motric Bio, and Mitsubishi Tanabe Pharma;

and intellectual property interest in KIM-1 and kidney organoid patents assigned to Mass General/Brigham. Wenxin Xu reports a consulting or advisory role for Eisai, Exelixis, Jazz Pharmaceuticals, Xencor, Merck, Aveo, Celdara, and Deciphera; and institutional research funding from OncoHost, Arsenal Biosciences, and Merck. Clara Steiner and Marc Machaalani have nothing to disclose.

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